

Thermodynamics

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17th March 2026

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1 Introduction

1.1 State Variables

These are variables which we can define for a thermodynamic system that describes the system at any point in time. The state variables for a system may change if we do different processes on it but if at a later time, the system were to be brought back to how it was before, then the state variables must also return to the same value as before.

The two common state variables that we can measure with our senses or using instruments are Pressure(P) and Volume(V). In the absence of any constraints, these two state variables can change independently of each other. That is, a gas can take any possible value of pressure at a particular value of volume, and vice versa. But when certain constraints are introduced, they lock the variables into a single function such that if I vary one, the other one also changes. We shall see one such constraint in the next section.

1.2 Quasi-Static process

It is a process that is so slow that the system is assumed to be close to (or at) equilibrium at each instant of time.

Need for a Quasi-static process: Equations of state that connect the different state variables together are only described when the system is in a state of equilibrium.

If processes happen at time-scales that are at least an order of magnitude larger than the relaxation times of the system, then the process can be assumed to be quasi-static.

All quasi-static processes are not reversible but all reversible process must be quasi-static.

2 The Zero'th Law of Thermodynamics

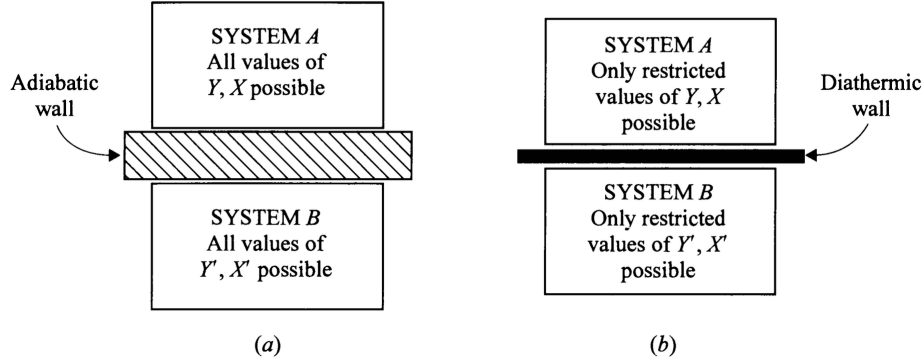
2.1 Thermal Equilibrium

In thermodynamics, an isolated system being in thermal equilibrium is introducing a constraint so as to make its state variables no longer independent of each other. That means I can write one of the variables as a function of another variables and there exist a function $f(P, V)$ such that

$$f(P, V) = 0$$

If I have two systems in thermal equilibrium but separated by an adiabatic wall, both the systems can independently be in thermal equilibrium. That is, $f(P_1, V_1) = f(P_2, V_2) = 0$. In other words, an equilibrium state of the first system can coexist with any equilibrium state of the second system.

But when the two systems are separated by a diathermic wall instead, and they attain thermal equilibrium, the state variables of the combined system will change spontaneously until an equilibrium state of the combined system has been attained. Since the four state variables are then related, I can write $f(P_1, V_1, P_2, V_2) = 0$



Thermodynamic Equilibrium A system which is simultaneously in mechanical, chemical and thermal equilibrium is said to be in thermodynamic equilibrium. There exists equations of state that govern a thermodynamics equilibrium.

2.2 The Zeroth Law

Definition: Say I have two systems — $\mathcal{A}$ & $\mathcal{B}$, separated by an adiabatic wall and are separately in thermal equilibrium with a third system $\mathcal{C}$. Then it is found that $\mathcal{A}$ and $\mathcal{B}$ are in thermal equilibrium with each other.

Say I'm able to define the state of $\mathcal{A}$, $\mathcal{B}$ and $\mathcal{C}$ by state variables P and V . Since $\mathcal{A}$ and $\mathcal{C}$ are in thermal equilibrium and are separated by a **diathermic wall**, the state variables of the combined system are not independent and there exist a function F_1 such that

$$F_1(P_A, V_A, P_C, V_C) = 0 \quad (1)$$

$$\implies P_C = f_1(P_A, V_A, V_C) \quad (2)$$

and similarly for $\mathcal{B}$ and $\mathcal{C}$,

$$F_2(P_B, V_B, P_C, V_C) = 0 \quad (3)$$

$$\implies P_C = f_2(P_B, V_B, V_C) \quad (4)$$

Similarly, since $\mathcal{A}$ is in thermal equilibrium with $\mathcal{B}$ according to zeroth law, There exist a function F_3 such that

$$F_3(P_A, V_A, P_B, V_B) = 0 \quad (5)$$

We see that equation (2) and (4) has V_C but (5) does not. That makes sense because by the zeroth law, the equilibrium between $\mathcal{A}$ and $\mathcal{B}$ does not depend on $\mathcal{C}$. Thus, if I were to equate (2) and (4) (since they both equal P_C), I need to write it in a form where V_C cancels from both sides. For this, I shall write f_1 and f_2 as

$$f_1 = \Phi(P_A, V_A) \cdot \zeta(V_C) + \eta(V_C)$$

$$f_2 = \Phi(P_B, V_B) \cdot \zeta(V_C) + \eta(V_C)$$

Since f_1 and f_2 are equal, I can write

$$\Phi(P_A, V_A) = \Phi(P_B, V_B)$$

Note that $\Phi(P_A, V_A) - \Phi(P_B, V_B) = 0$ is simply our function $F_3(P_A, V_A, P_B, V_B)$ according to (5)

If we call the scalar value of this function as “*empirical temperature*”, we see that any two systems at thermal equilibrium with each other have the same value of this quantity called temperature. Thus, I can write the same for system C too.

$$\boxed{\Phi(P_A, V_A) = \Phi(P_B, V_B) = \Phi(P_C, V_C) = \theta}$$

Thus the zeroth law implies the existence of a property called temperature which has the same value across systems that are at thermal equilibrium. This is the working principle behind all thermometers.

2.3 Thermometry

Some common thermometers are constant volume gas thermometers, resistance thermometers, magnetization thermometers, linear/area/volume expansion thermometers. All this thermometers have one thing in common – a thermodynamic property X that varies linearly with temperature, I can write this as $X(\theta)$ but it is more useful to write the temperature in terms of this thermodynamic property as

$$\theta(X) = aX \tag{6}$$

where a is some constant.

Two fixed points are required to define a temperature scale. Celsius scale is based on freezing and boiling point of water. But it was discarded from being a standard for the difficulty in reproducing the standard points as the temperature at which water freezes/boils changes with pressure.

Later, the Kelvin scale was introduced. For this scale one fixed point was the triple point of water (which happens at only one particular temperature and pressure) and the other fixed point is the theoretical minimum temperature a body could have. The latter was called 0 Kelvin and the former is 273.16 Kelvin.

Lets say I know the thermodynamic property of a material at the triple point temperature X_{TP} . I can then substitute $273.16K$ for $\theta(X_{TP})$ and obtain an expression for the constant a as

$$a = \frac{273.16 \text{ K}}{X_{TP}}$$

Substituting this in (6), I get

$$\theta(X) = \left(\frac{273.16 \text{ [K]}}{X_{TP}} \right) X \tag{7}$$

2.4 Equations of State

A system in thermal equilibrium do not have all independent variable and one arrives at a condition like the one obtained before — $\Phi(P, V) = \theta$. Some other common equations of state are

1. Ideal Gas Law :

$$PV - nRT = 0$$

2. Van-der Waal's Equation :

$$\left(P + a \frac{n^2}{V^2}\right) (V - nb) - nRT = 0$$

where a is related to the molecular interactions and b is related to the finite size of the particles both of which the ideal gas law ignored.

3. Virial Equation:

$$P - \frac{Nk_B T}{V} \left[1 + \frac{N}{V} B_2(T) + \frac{N^2}{V^2} B_3(T) + \dots \right] = 0$$

where $B_2(T), B_3(T), \dots$ are called virial coefficients.

4. Curie's Equation of state of paramagnet

$$M - \frac{CH}{T} = 0$$

$M \longrightarrow$ Magnetization

$H \longrightarrow$ Applied external magnetic field

$C \longrightarrow$ Curie constant

$T \longrightarrow$ Temperature

Given an equation of state, it can be solved for one variable in terms of the other variables like say volume expressed in terms of pressure and temperature— $V(P, T)$. Thermodynamic equilibrium ensures that infinitesimal changes in state variables doesn't change the equilibrium.

By theorem of partial differentials, if z is an exact differential of x and y , it can be written as

$$dz = \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy$$

If it can't be written in this form, it is called a non-exact differential equation.

P, V and T represent actual variables of state in thermodynamic equilibrium and their differentials are exact differentials.

Writing $V(P, T)$

$$dV = \left(\frac{\partial V}{\partial T} \right)_P dT + \left(\frac{\partial V}{\partial P} \right)_T dP \quad (8)$$

The term $\left(\frac{\partial V}{\partial T} \right)_P$ is simply the volume thermal expansion and I can express it as

$$\begin{aligned} dV &= \beta V dT \\ \Rightarrow \frac{\partial V}{\partial T} &= \beta V \end{aligned}$$

Similarly, $\left(\frac{\partial V}{\partial P} \right)_T$ is the negative of inverse of bulk modulus (or compressibility $-\kappa$), with a volume term also, to make it volumetric strain instead of dV .

$$\begin{aligned} \kappa &= -\frac{dV/V}{dP} \\ \Rightarrow \left(\frac{\partial V}{\partial P} \right)_T &= -\kappa V \end{aligned}$$

Thus, I can write (8) as

$$dV = \beta V dT - \kappa V dP$$

Similarly, I can write Pressure and Temperature as

$$dP = \left(\frac{\partial P}{\partial T} \right)_V dT + \left(\frac{\partial P}{\partial V} \right)_T dV \quad (9)$$

$$dT = \left(\frac{\partial T}{\partial P} \right)_V dP + \left(\frac{\partial T}{\partial V} \right)_P dV \quad (10)$$

Theorems on Partial Differential Equations

The following equations are built on variables x, y and z that have a fundamental relationship of the kind $f(x, y, z) = 0$

1. Reciprocal Theorem

$$\left(\frac{\partial y}{\partial x} \right)_z = \frac{1}{\left(\frac{\partial x}{\partial y} \right)_z} \quad (11)$$

2. Cyclic theorem on triple scalar product

$$\left(\frac{\partial x}{\partial y} \right)_z \left(\frac{\partial y}{\partial z} \right)_x \left(\frac{\partial z}{\partial x} \right)_y = -1 \quad (12)$$

This might seem counter-intuitive at first, but here's the proof

proof: Let there exist x , y and z such that $\exists f(x, y, z)$. I can write them in exact differential form as follows

$$dz = \left(\frac{\partial z}{\partial x} \right)_y dx + \left(\frac{\partial z}{\partial y} \right)_x dy \quad (13)$$

Similarly, I can write dx as

$$dx = \left(\frac{\partial x}{\partial y} \right)_z dy + \left(\frac{\partial x}{\partial z} \right)_y dz$$

Substituting (13) in above, we get

$$\begin{aligned} dx &= \left(\frac{\partial x}{\partial y} \right)_z dy + \left(\frac{\partial x}{\partial z} \right)_y \left[\left(\frac{\partial z}{\partial x} \right)_y dx + \left(\frac{\partial z}{\partial y} \right)_x dy \right] \\ \Rightarrow dx &= \left(\frac{\partial x}{\partial y} \right)_z dy + dx + \left(\frac{\partial x}{\partial z} \right)_y \left(\frac{\partial z}{\partial y} \right)_x dy \\ \Rightarrow \boxed{\left(\frac{\partial x}{\partial y} \right)_z \left(\frac{\partial y}{\partial z} \right)_x \left(\frac{\partial z}{\partial x} \right)_y = -1} \end{aligned}$$

3. **Test for Exactness:** Suppose I have a differential equation of the form

$$df = M(x, y)dx + N(x, y)dy \quad (14)$$

It is exact iff

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

proof: Since (14) is exact, I can compare it to (13) and write

$$M = \left(\frac{\partial f}{\partial x} \right)_y \quad \text{and} \quad N = \left(\frac{\partial f}{\partial y} \right)_x$$

Then, taking a partial derivative of M with respect of y and N with respect of x , we get

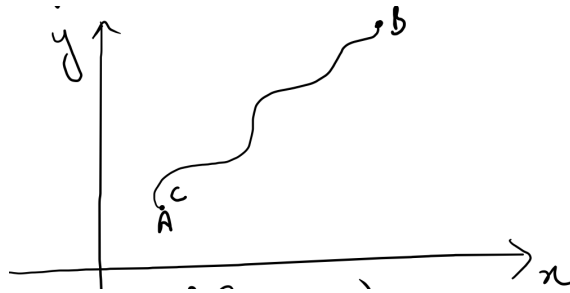
$$\frac{\partial M}{\partial y} = \frac{\partial^2 f}{\partial y \partial x} \quad \text{and} \quad \frac{\partial N}{\partial x} = \frac{\partial^2 f}{\partial x \partial y}$$

If a function is smooth and continuous, the second derivative would commute

$$\begin{aligned} \Rightarrow \frac{\partial^2 f}{\partial y \partial x} &= \frac{\partial^2 f}{\partial x \partial y} \\ \Rightarrow \frac{\partial M}{\partial y} &= \frac{\partial N}{\partial x} \end{aligned}$$

4. **Path independence:** An exact differential is path independent.

proof: Say a system goes from state-A to state-B through some path-C. we get



$$\int_A^B df(x, y) = \int_A^B \left(\frac{\partial f(x, y)}{\partial x} \right)_y dx + \int_A^B \left(\frac{\partial f(x, y)}{\partial y} \right)_x dy$$

parameterizing x and y in terms of a single variable t , since $dx = \frac{dx}{dt} dt$ and $dy = \frac{dy}{dt} dt$, we get...

$$\begin{aligned} \int_{t_A}^{t_B} df(x, y) &= \int_{t_A}^{t_B} \left(\frac{\partial f}{\partial x} \right) \frac{dx}{dt} dt + \int_{t_A}^{t_B} \left(\frac{\partial f}{\partial y} \right) \frac{dy}{dt} dt \\ &= \int_{t_A}^{t_B} \left(\frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} \right) dt \\ &= \int_{t_A}^{t_B} \left(\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy \right) \\ &= \int_{t_A}^{t_B} df(x(t), y(t)) = f(t_B) - f(t_A) \end{aligned}$$

This means that the value of integral only depends on the initial and final points and not the path taken between the two states ie, path independence.

3 The First Law of Thermodynamics

3.1 Internal Energy

Refers to the sum total of all energies of all particles that make up the system. It includes kinetic energy, vibrational energy, intermolecular potential energy, etc. For an ideal gas, we ignore the intermolecular attraction between the particles of the system and also for polyatomic species, we assume there is no vibration of bonds. In such a scenario, the Internal energy is given by the sum of the kinetic energies of the particles and therefore is only dependent on the temperature of the system.

$$U = \frac{f}{2} nRT$$

where f is the degrees of freedom for a molecule. At low temperatures where molecules don't vibrate,

Monoatomic : $f = 3$ (3 translational modes)

Diatomic : $f = 5$ (3 translational modes + 2 rotational modes)

Polyatomic : $f = 6$ (3 translational modes + 3 rotational modes)

The First Law

Finding the absolute value of the Internal Energy of a system is not merely as useful as finding the change in Internal Energy during a process. According to the first law of thermodynamics, there are only two ways of changing the internal energy of a system. They are by transfer of heat and by doing Work. That is

$$\Delta U = Q + W_{\text{ext}}$$

3.2 Doing Work on a gas

When we say that we are doing work on a gas, it implies that we are exerting external force (more precisely pressure) on a gas to decrease the volume occupied by the gas system. Similarly, when the gas does work, it pushes against an external force to expand its volume.

$$dW_{\text{ext}} = \vec{F} \cdot \vec{ds}$$

$$dW_{\text{ext}} = (P_{\text{ext}} \vec{A}) \cdot \vec{ds}$$

During a compression or expansion of a volume, the surface that encapsulates that volume moves inwards or outwards therefore, the angle between Area element and displacement element is 0° . Hence, I can write

$$dW_{\text{ext}} = P_{\text{ext}} (A \cdot ds)$$

$$dW_{\text{ext}} = P_{\text{ext}} \cdot dV$$

The derivation only yields us the expression for work done, it is upto to us to make the sign convention consistent with the first law. What I want is W_{ext} to be positive when an external force compresses the system and to be negative when the gas does work against an external force to expand its volume. Therefore, if V denotes the volume of the the gas, I want $W_{\text{ext}} > 0$ when $dV < 0$. After correction, I get

$$dW = -P_{\text{ext}} dV$$

$$W = - \int_{V_1}^{V_2} P_{\text{ext}} dV$$

Graphically, W_{ext} is given by the area under P - V graph.

Reversible and Irreversible Work

There are mainly two types of works - Reversible Work and Irreversible Work. Compression or expansion that involves a finite number of steps do Irreversible work on the system. This is because, if we compress a gas irreversibly, then let it expand, it won't return to the starting state of (P, V, T) .

Meanwhile, reversible work involves doing a process infinitely slowly so that at the end of each step, the state of the gas is only infinitesimally different than before and the gas can return to its starting state if I allow it to. Also, during a reversible process, the gas is in equilibrium with the surrounding at every instant.

$$\implies P_{\text{int}} = P_{\text{ext}}$$

It can be seen from the P - V graph that, for the same change in volume, work done on the gas is minimum for reversible compression and maximum for a single step irreversible compression. Similarly, work done by the gas is maximum for reversible expansion and minimum for a single step irreversible expansion.

4 Thermodynamic Processes

4.1 Isothermal Process

In an Isothermal process, Temperature of the system remains constant. That is, $\Delta T = 0$. Now since, the Internal energy U of a system is only function of the temperature, I can write

$$\Delta U_{\text{isothermal}} = 0$$

According to Ideal Gas law, $PV = nRT \implies PV = \text{constant}$, for Isothermal process.

To obtain expression for Work done on the gas during reversible isothermal process, I use

$$dW = -P_{\text{ext}} dV$$

$$W = - \int_{V_1}^{V_2} P_{\text{ext}} dV$$

For reversible processes, $P_{\text{ext}} = P_{\text{int}}$. And I can replace P_{int} with an expression in V using ideal gas law.

$$W = - \int_{V_1}^{V_2} \frac{nRT}{V} dV$$

$$W = -nRT \int_{V_1}^{V_2} \frac{1}{V} dV$$

$$W_{\text{isothermal}} = nRT \ln \left(\frac{V_1}{V_2} \right)$$

Now, since $\Delta U = 0 \implies W_{\text{ext}} = -Q$. Thus, the above equations holds for finding heat transferred as well.

Suppose the isothermal compressibility κ of substance is known to us and we have to derive an expression for work done when a certain amount of pressure is applied to our body. We know that

$$\kappa = \frac{-1}{V} \left(\frac{\partial V}{\partial P} \right)_T$$

$$\implies dV = -\kappa V dP$$

Substituting this for dV in the equation for work $dW = -P_{\text{ext}} dV$

$$\int dW = \int_{P_1}^{P_2} \kappa V P_{\text{ext}} dP$$

We treat V to be constant, because a finite change in pressure can be produced by very small values of volume change. Thus,

$$W = \frac{\kappa V}{2} (P_2^2 - P_1^2)$$

4.2 Adiabatic Process

In an adiabatic process, there is no heat exchange between the system and the surrounding. That is, $dQ = 0$. According to first law of thermodynamics,

$$dU = dQ + dW$$

substituting $dQ = 0$, we get

$$dU = dW_{\text{adia}} = -P_{\text{ext}} dV$$

for an Isochoric process, $dU = dQ_v = n c_v dT$. Now when I observe the equation $dU = n c_v dT$, I see that all terms are state functions, therefore although it has been derived for a specific

kind of process (Isochoric), it is applicable to all processes. Substituting that into dU in above equation, I get,

$$n c_v dT = -P_{\text{ext}} dV$$

Here, although I've used P_{ext} , for every quasi-static reversible process, $P_{\text{ext}} = P_{\text{int}}$. Replacing it in the above equation and using ideal gas law to write P_{int} in terms of V , T and n .

$$n c_v dT = -\frac{nRT}{V} dV$$

$$\begin{aligned} \frac{c_v}{RT} dT &= -\frac{1}{V} dV \\ \int_{T_1}^{T_2} \frac{c_v}{RT} dT &= -\int_{V_1}^{V_2} \frac{1}{V} dV \end{aligned}$$

I assume that c_v is constant throughout the temperature range

$$\frac{c_v}{R} \int_{T_1}^{T_2} \frac{1}{T} dT = \int_{V_2}^{V_1} \frac{1}{V} dV$$

$$\frac{c_v}{R} \ln \left(\frac{T_2}{T_1} \right) = \ln \left(\frac{V_1}{V_2} \right)$$

$$\left(\frac{T_2}{T_1} \right)^{\frac{c_v}{R}} = \left(\frac{V_1}{V_2} \right)$$

$$\left(\frac{P_2 V_2}{P_1 V_1} \right)^{\frac{c_v}{R}} = \left(\frac{V_1}{V_2} \right)$$

$$\left(\frac{P_2}{P_1} \right)^{c_v} = \left(\frac{V_1}{V_2} \right)^{c_v + R}$$

$$\left(\frac{P_2}{P_1} \right) = \left(\frac{V_1}{V_2} \right)^{\frac{c_p}{c_v}}$$

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

Therefore, for an adiabatic process, $PV^\gamma = \text{constant}$. Where, $\gamma = \frac{c_p}{c_v}$. It is also worth noting that $\gamma = 1 + \frac{2}{f}$ where f is the degrees of freedom for the molecule.

Adiabatic Work

Work done on the gas during a reversible (quasi-static) adiabatic process is path independent and thus, I can write it as

$$dW = -P_{\text{ext}} dV = -P_{\text{int}} dV$$

for an adiabatic process $P_{\text{int}} V^\gamma = \text{constant}$ (say k)s

$$W = \int_{V_i}^{V_f} \frac{k}{V^\gamma} dV$$

$$W = k \left[\frac{V^{-\gamma+1}}{-\gamma+1} \right]_{V_i}^{V_f}$$

$$W = \left[\frac{k V_f^{-\gamma+1}}{-\gamma+1} - \frac{k V_i^{-\gamma+1}}{-\gamma+1} \right]$$

substitute $k = P_i V_i^\gamma = P_f V_f^\gamma$

$$W = \left[\frac{(P_f V_f^\gamma) V_f^{-\gamma+1}}{-\gamma+1} - \frac{(P_i V_i^\gamma) V_i^{-\gamma+1}}{-\gamma+1} \right]$$

$$W = \left[\frac{(P_f V_f^\gamma) V_f^{-\gamma+1} - (P_i V_i^\gamma) V_i^{-\gamma+1}}{1-\gamma} \right]$$

$$W = \frac{P_f V_f - P_i V_i}{1-\gamma} = \frac{nRT_f - nRT_i}{1-\gamma}$$

Therefore, work done on the gas during an adiabatic process is given by

$$W_{\text{adiabatic}} = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1} = \frac{nR(T_1 - T_2)}{\gamma - 1}$$

5 Enthalpy

Definition : By definition, enthalpy is the heat content of the system. Physically it gives the total energy of a system to exist in this universe, to push against an external atmospheric pressure and claim its rightful volume. Mathematically,

$$H = U + PV$$

Change in enthalpy : Just like internal energy - U , knowing its change during a process is more important than its raw value at any instant.

$$\begin{aligned}dH &= dU + d(P_{\text{int}}V) \\dH &= (dQ + dW) + (P_{\text{int}} \cdot dV + V \cdot dP_{\text{int}}) \\dH &= (dQ - P_{\text{ext}} dV) + (P_{\text{int}} dV + V dP_{\text{int}})\end{aligned}$$

At equilibrium state, when our system has expanded to match the external pressure, $P_{\text{int}} = P_{\text{ext}}$. Therefore, I can write

$$\begin{aligned}dH &= (dQ - P_{\text{ext}} dV) + (P_{\text{ext}} dV + V dP_{\text{ext}}) \\dH &= dQ + V dP_{\text{ext}}\end{aligned}$$

when pressure is constant, enthalpy equals heat added. That is, $dH = dQ_p$

Why enthalpy? Most reactions happen exposed to the atmosphere, even if the volume of reactants change during the reaction, the pressure it exerts outwards always match the atmospheric pressure. In this case, it is easier to deal with Enthalpy when talking about heat of reactions.

6 Entropy & The second law

6.1 General definition

Entropy(S) is a state function that gives the measure of disorder in a system. A highly ordered system has lower entropy than a highly disordered one. Speaking from a microscopic point of view, it is determined by the number of Microstates that a system can have.

Macrostate : Refers to the properties of the system that can be measured from a macroscopic point of view, like temperature, pressure, volume etc.

Microstate : Refers to the properties of the particles making up the system. Like exact position (x , y and z) and exact momentum (p_x , p_y and p_z) of each particle. That is, each particles microstate can be uniquely described by a single point on a 6-D cartesian system with the axes - x , y , z , p_x , p_y and p_z

6.2 Statistical Definition

Statistically, Entropy can be defined using Boltzmann's definition

$$S = k_B \ln(W)$$

Where, W is the number of available Microstates.

6.3 Change in Entropy

Although getting the exact value of entropy at any given moment requires the full Microstate description of a system, change in entropy can be found using macroscopic parameters also. Mathematically, change in entropy for **any reversible** process is given by

$$dS = \frac{dQ_{\text{rev}}}{T}$$

Below given list contains the expression for calculating change in entropy for some simple processes. All these can be derived from above equation.

Type of process	Change in Entropy
Cyclic Process	$\Delta S = 0$
Reversible Adiabatic	$\Delta S = 0$
Reversible Phase Change	$\Delta S = \frac{\Delta H}{T}$
Reversible Isobaric Change	$\Delta S = n c_p \ln \left(\frac{T_f}{T_i} \right)$

It might seem counter-intuitive for the change in entropy for reversible adiabatic process to always be 0 because, if I let a gas undergo adiabatic expansion, the number of possible positional micro-states that the gas particles can occupy increases (because of the increased volume). It seems like Entropy should increase. But the catch here is that when the gas undergoes such an expansion, it cools down and therefore the range of available velocity (and momenta) shrinks. This reduces the number of available momentum Microstates. Both these effects perfectly cancel out leading to a net 0 change in entropy.

6.4 Clausius Inequality

Imagine a system that goes from a state (P_1, V_1, T_1) to a state (P_2, V_2, T_2) through two pathways - one reversible and one irreversible. Although paths are different, the change in state functions - ΔU , ΔS etc, are the same for both pathways.

$$dU = dQ_{\text{rev}} - P_{\text{int}} dV \quad (15)$$

$$dU = dQ_{\text{irrev}} - P_{\text{ext}} dV \quad (16)$$

subtracting both equations, I get...

$$dQ_{\text{rev}} - dQ_{\text{irrev}} = (P_{\text{int}} - P_{\text{ext}}) dV$$

Now, if I observe the RHS of the equation, it must always be greater than zero. Because if $P_{\text{int}} > P_{\text{ext}}$, expansion happens. ie, $dV > 0$. $\implies$ product is positive. And also if $P_{\text{int}} < P_{\text{ext}}$, compression happens. ie, $dV < 0$. $\implies$ product is again positive.

$$\begin{aligned}
(P_{\text{int}} - P_{\text{ext}}) dV &> 0 \\
\implies dQ_{\text{rev}} - dQ_{\text{irrev}} &> 0 \\
dQ_{\text{rev}} &> dQ_{\text{irrev}}
\end{aligned}$$

Now, since change in entropy of the system during this change is given by $dS = \frac{dQ_{\text{rev}}}{T_{\text{rev}}}$, I can replace dQ_{rev} in terms of dS .

For a reversible process, $T_{\text{rev}} = T_{\text{surr}}$ since a non-adiabatic system is in thermal equilibrium with the surrounding at all instants. Note that this entropy change refers to the change in entropy as the system goes from the initial state to final state and is applicable to both the reversible and irreversible paths.

$$\begin{aligned}
T_{\text{surr}} dS &> dQ_{\text{irrev}} \\
dS &> \frac{dQ_{\text{irrev}}}{T_{\text{surr}}}
\end{aligned}$$

Now, I integrate both sides over an entire cycle. ie, take the closed loop integral.

$$\oint dS > \oint \frac{dQ_{\text{irrev}}}{T_{\text{surr}}}$$

Since, entropy is a state function, its integral over a complete loop must be zero. That is,

$$\boxed{\oint \frac{dQ_{\text{irrev}}}{T_{\text{surr}}} < 0}$$

This is the Clausius Inequality. An important thing to note here is why the temperature used here is T_{surr} and not T_{sys}

Since the Surrounding acts as a temperature reservoir and are internally reversible, any process that happens to the Surrounding can be considered reversible. Therefore, the expression for change in entropy for the surrounding is not only for heat added in reversible processes but all processes.

$$\Delta S_{\text{surr}} = \frac{Q}{T_{\text{surr}}}$$

In this case, $dQ_{\text{surr}} = dQ_{\text{irrev}}$ of the system. Therefore, during one full cycle of the system, the change in entropy of the surrounding is given by

$$\Delta S_{\text{surr}} = \oint \frac{dQ_{\text{surr}}}{T_{\text{surr}}} = \oint \frac{-dQ_{\text{irrev}}}{T_{\text{surr}}}$$

6.5 The Second Law of Thermodynamics

It is clear from above that ΔS_{surr} can be written as

$$\Delta S_{\text{surr}} = \oint \frac{dQ_{\text{irrev}}}{T_{\text{surr}}}$$

and according to Clausius Inequality, this expression has to be greater than 0.

If I now introduce another term $\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$. I see that for **all reversible processes**, $\Delta S_{\text{univ}} = 0$ and for **all irreversible processes**, $\Delta S_{\text{univ}} > 0$

To understand why this works, let's think of a common scenario, I have a hot closed cup of tea at $T_{\text{cup}} = 60^\circ\text{C}$ exposed to the surrounding at $T_{\text{surr}} = 20^\circ\text{C}$. What happens now is pretty obvious, the tea loses heat through the cup walls until it reaches thermal equilibrium with the surrounding. Let Q be the magnitude of heat that is transferred.

$$\Delta S_{\text{sys}} = -\frac{Q}{T_{\text{cup}}}$$

$$\Delta S_{\text{surr}} = \frac{Q}{T_{\text{surr}}}$$

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$$

$$\Delta S_{\text{univ}} = -\frac{Q}{T_{\text{cup}}} + \frac{Q}{T_{\text{surr}}}$$

$$\Delta S_{\text{univ}} = Q \left(\frac{1}{T_{\text{surr}}} - \frac{1}{T_{\text{cup}}} \right)$$

since $T_{\text{cup}} > T_{\text{surr}}$,

$$\left(\frac{1}{T_{\text{surr}}} - \frac{1}{T_{\text{cup}}} \right) > 0$$

$$\Delta S_{\text{univ}} > 0$$

This also explains why heat wouldn't flow from a colder body to a hotter body. Although it is allowed by the First Law, It would imply that $\Delta S_{\text{univ}} < 0$. That would violate the Second Law.